MATHEMATICS

University *of* NORTH ALABAMA

Spring 2025 Math Problem of the Week: Problem #1

Due noon Tuesday, January 21
Open to all UNA undergraduates

Directions: Each Monday, a new problem will be posted on the bulletin board near the front desk and around the 3rd floor of CPTMTH. You have until noon of the proceeding Monday to deliver your written solution to me (Dr. Bryant) in CPTMTH 328 or, if you cannot find me, you may leave it with Ms. Buckley in CPTMTH 377. All work must be your own. You are required to justify your response fully. If you utilize a well-known theorem, state the theorem's full name and proposition, showing that the conditions of the theorem are met. Good luck!

Problem #1: Find all triples of natural numbers (a, b, n), with $n > 1$, that satisfy the relation $a^n + b^n = 2025$. Show that your solution set is complete.